

IS 4513 001; IS 6433 001

Industrial Control Systems Security ; Supervisory Control and Data Acquisition

Course Syllabus - Spring 2025



Course Information

Course Description: Cross-listed courses include IS 4513.001, IS 6433.001

Credit Hours: 3; 3

Course Modality: Traditional in-person

Meeting Times

Duration: 01/21/2025-05/16/2025; 01/21/2025-05/16/2025

Campus: Main Campus; Main Campus

Location: [BB 3.01.08](#); [BB 3.01.08](#)

Time(s): MW 6:00 PM - 7:15 PM; MW 6:00 PM - 7:15 PM

Learning Goals

At the end of the course, you will be able to:

- Understand the Components of ICS and SCADA Systems
Outcome
 - Identify and describe the main components of Industrial Control Systems (ICS), including Supervisory Control and Data Acquisition (SCADA) systems, Programmable Logic Controllers (PLCs), and Human-Machine Interfaces (HMIs).
 - Explain the differences between IT and OT environments and their unique security challenges.
- Understand ICS-Specific Protocols and Architectures
Outcome

- Describe common ICS communication protocols such as Modbus, DNP3, OPC-UA, and IEC 61850. Analyze ICS network architectures, including the Purdue Model, and identify common entry points for cyber threats.
- Explain the Role of ICS in Critical Infrastructure

Outcome

 - Understand the importance of ICS in sectors such as energy, manufacturing, water treatment, and transportation.
 - Assess the potential impacts of ICS cybersecurity incidents on public safety, operational continuity, and national security.
- Recognize Common ICS Cyber Threats and Vulnerabilities:

Outcome

 - Identify common threats such as ransomware, insider threats, Advanced Persistent Threats (APTs), and supply chain attacks targeting ICS.
 - Assess vulnerabilities unique to ICS, such as insecure legacy devices, weak authentication mechanisms, and unpatched software.
- Analyze ICS Network Traffic:

Outcome

 - Use tools such as Wireshark or tcpdump to capture and analyze traffic from ICS protocols like Modbus and DNP3.
 - Identify anomalies and potential security incidents in ICS network traffic.
- Configure and Monitor Security Tools for ICS:

Outcome

 - Deploy and configure Intrusion Detection Systems (IDS) like Snort or Suricata to monitor ICS networks.
 - Develop intrusion detection rules tailored for ICS protocols and devices.
- Simulate and Respond to ICS Cyber Incidents:

Outcome

 - Conduct hands-on exercises in simulating attacks on ICS environments, such as unauthorized PLC command execution or malware propagation.

- Develop and execute incident response plans specific to ICS environments, including root cause analysis and recovery.

Communicate with Me

Instructor Name: Ian Burres

Department

Information Systems and Cyber Security

Office Location

NPB 3.226

Student Hours

M/W 1:30 PM - 3:30 PM

Phone Number(s): 954.415.9538

Email Address: ian.burres@utsa.edu

Preferred Method of Communication

Email

About Me & My Teaching Philosophy

My name is Ian Burres, and I am excited to be part of your academic journey this semester.

I earned my master's in computer science from the University of Illinois in 2015. Before joining UTSA, I worked with Sandia National Laboratories as a Senior cybersecurity Research and Development Engineer. I also taught as an adjunct professor at the University of New Mexico and as a full-time tenure track professor at Palm Beach State College, located in Boca Raton, FL.

My professional background includes digital forensics, malware analysis and reverse engineering, industrial control systems, software engineering, and vulnerability assessments. I am also currently exploring research in the area of machine learning, so if you are interested in that field, please feel free to ask.

Course Materials

NA

Subtitle: NA

ISBN: NA

Additional Information

I will provide you with all of the information, handouts, and lab equipment needed to successfully complete this course.

Additional Course Information

This is a hands-on class that emphasizes the concept of experiential learning. As a result, we will spend a lot of time working with hardware in the SCADA lab and with virtual environments, which I will help you install and setup.

You do not need a textbook for this course.

Assessments and Assignments

Assignments:

4 Lab Projects

10 Short Certification Modules through CISA (these are optional and will count as extra credit).
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Assessments:

2 In-Class Exams

4 Hands-On Assessments

A NOTE ON LATE ASSIGNMENT SUBMISSIONS: I will accept late submissions up to 4 days past the due date. However, I will apply a 10% PENALTY each day the assignment is late. That means if you submit something more than 4 days after it is due, you will receive a 0 for the assignment. I will not provide exceptions to this policy unless you have documented proof that you were unable to complete the work during this period. The other exceptions are jury duty or military obligations (I will need to see a copy of your orders though).

Activities and Grading

Activity	Quantity	%
Lab Projects	4	40
Total		100%

Activity	Quantity	%
Hands-On Assessments	4	40
Exams	2	20
Total		100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A+	98.0 – 100
A	92.0 - 97.9
A-	90.0 - 91.9
B+	88.0 - 89.9
B	82.0 - 87.9
B-	80.0 - 81.9
C+	78.0 - 79.9
C	72.0 - 77.9
C-	70.0 - 71.9
D+	68.0 - 69.9
D	62.0 - 67.9
D-	60.0 - 61.9
F	< 60.0

Grade Distribution and Letter Grade

Guidelines for the Use of Generative Artificial Intelligence

I encourage students to explore the use of generative artificial intelligence (AI) tools, such as ChatGPT, for all assignments and assessments. Any such use must be appropriately acknowledged and cited, following the guidelines established by the APA Style Guide, including the specific version of the tool used. Submitted work should include the exact prompt used to generate the content as well as the AI's full response in an Appendix. Because AI generated content is not necessarily accurate or appropriate, it is each student's responsibility to assess the validity and applicability of any generative AI output that is submitted. You may not earn full credit if inaccurate, invalid, or inappropriate information is found in your work. Deviations from these

guidelines will be considered violations of CMU's academic integrity policy. Note that expectations for "plagiarism, cheating, and acceptable assistance" on student work may vary across your courses and instructors. Please email me if you have questions regarding what is permissible and not for a particular course or assignment.

Course Expectations & Policies

Instructor-Initiated Drops

This course uses instructor-initiated drops for students who exceed the absence and/or missed assignment limit. Therefore, up to the last day for students to withdraw from an individual course, **March 21st**, you will be dropped **for failing to submit 2 Lab projects or for having 4+ unexcused absences**. Students will receive at least one courtesy warning when approaching the absence/missed assignment limit. Notification will be sent using ASAP to the student's email address. A subsequent absence or missed assignment will result in being dropped from the course. Notification of being dropped will also be sent through ASAP to the student's email address. *This drop does not affect enrollment in other courses. Please consult the [Dropping Courses webpage](#) for further details on the process and appeals.*

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law. For more information on your privacy and class recordings, review [Student Privacy \(FERPA\) in Virtual Classrooms and Other Educational Recordings](#) and the [Guide to Secure Video Conferencing Tools](#).

Academic Freedom

Academic freedom¹ is a cornerstone of the University. Academic freedom in its teaching aspect is fundamental for the protection of the rights of the teacher in teaching and of the student to freedom in learning.² Each faculty member is entitled to full freedom in the classroom discussing the subject that the faculty member teaches.³ The University of Texas at San Antonio will not penalize or discipline members of the faculty because of their exercise of academic freedom.

Along with this freedom comes responsibility. It is the responsibility of faculty members to ensure that topics taught are related to the classroom subject. Students should be free to take reasoned exception to the data or views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.⁴ It is not the proper role of the university or any outside agency to attempt to shield individuals from ideas and opinions they find unwelcome,

disagreeable, or even deeply offensive.⁵ Engaging with new ideas and perspectives helps students grow intellectually and is beneficial to the educational process.

1. Statement adapted from Texas A&M University's [Syllabus Statement Regarding Academic Freedom](#)
2. 1940 Statement of Principles on Academic Freedom and Tenure
3. Board of Regents Rule 31004; HOP Policy 4.02
4. American Association of University Professors Joint Statement on Rights and Freedoms of Students
5. The Chicago Statement

Our Commitment to Inclusivity

The University of Texas at San Antonio, a Hispanic Serving Institution situated in a global city that has been a crossroads of peoples and cultures for centuries, values diversity and inclusion in all aspects of university life. As an institution expressly founded to advance the education of Mexican Americans and other underserved communities, our university is committed to promoting access for all. UTSA, a premier public research university, fosters academic excellence through a community of dialogue, discovery and innovation that embraces the uniqueness of each voice.

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UTSA provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit www.utsa.edu/disability or call (210) 458-4157.
- Students at UTSA are responsible for ensuring their work is consistent with UTSA's standards for academic integrity. Students should review [Section 203 of the UTSA Student Code of Conduct](#) for

appropriate standards of academic integrity.

- UTSA provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UTSA Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UTSA.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.